

Answer any 2

1. a) What is preference based approach of consumer's behavior? Explain and interpret the assumptions of preference based approach.
b) Define indifference set. Examine that if a preference relation satisfies strong monotonicity, its indifference curves must be downward sloping.
2. a) What are the twin definitions of convexity? In what respect they are different from strong convexity-examine graphically?
b) Show that,

Convexity of preferences $\Leftrightarrow$ USC(x) is convex $\Leftrightarrow$ $u(\cdot)$ is quasiconcave

- c) Prove that, if a preference relation satisfies monotonicity and continuity, then there exist a utility function $u(\cdot)$ representing such preference relation.
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3. Consider the Cobb-Douglas utility function:
$$u(x_1, x_2) = Kx_1^\alpha x_2^{1-\alpha} \quad \text{for some } \alpha \in (0,1) \text{ and } K > 0$$
 - i) Derive the Walrasian demand functions and examine their properties.
 - ii) How do you get the indirect utility functions? Explain their properties and also verify Roy's identity.
 - iii) Solve the expenditure minimization problem and examine the characteristics of the compensated demand function.
 4. a) Define utility function. Explain the desirability of consumer's preference by the axiom of monotonicity and strong monotonicity.
b) Show that,

Strong Monotonicity Monotonicity LNS

*X*c) Examine the following properties of preference relations:

- i) Homogeneity
- ii) Homotheticity

Answer any one (1) of the

Time: 50 minutes

1. a) Distinguish between compensating variation (CV) and equivalent variation (EV).
In what respect they are different from the consumer's surplus (CS)
b) Illustrate that for a i) decrease in price of a commodity (when both goods are normal)

$$EV > CS > CV$$

ii) for price increase $EV < CS < CV$

2. a) Define production function, production set, transformation function and transformation frontier.
b) Graphically explain the properties of the production set.
3. a) Define profit function. If the production set Y is closed and have free disposal, then what properties does the profit function satisfies?

b) Consider a consumer with the utility function: $u(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{2}}$

Consider a) $\{p^0, m^0\} = \{4, 4, 300\}$

b) $\{p^1, m^0\} = \{2, 4, 300\}$

Compute the indirect utilities and costs.

Calculate EV, CV, and CS. Based on their relationship, examine the nature of the commodities.

4. a) Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = R_+^N$, then examine the properties satisfied by the Walrasian demand function $x(p, w)$. In what situations the sufficient condition of UMP be violated?

b) Examine the following properties of preference relations:

- i) Homogeneity
- ii) Homotheticity

Tutorial exam 03

Time: 50 minutes; Marks 20

Consider a continuous-time Ramsey-Cass-Koopmans Model in which aggregate output is produced according to the production function $Y(t) = F(K(t), A(t)L(t))$; where technology $A(t)$ grows exogenously and labor $L(t)$ grows at rate n . The real interest rate is $r(t)$. The representative household maximizes the discounted stream of utility given by

$$U = \int_0^{\infty} e^{-\rho t} u(c(t)) \frac{L(t)}{H} dt$$

where the instantaneous utility function follows constant relative risk aversion (CRRA).

- a) i) Describe the differential equations governing $\dot{k}$ and $\dot{c}$. [6]
ii) Use them to determine the $\dot{k}=0$ and $\dot{c}=0$ loci.
iii) Illustrate and interpret the phase diagram.
- b) Briefly explain the socially optimal level of capital and the Golden Rule level of capital. [2]
- c) Estimate the speed of adjustment through a Taylor rule approximation. Determine the value of μ_1 (the speed of adjustment) assuming values of independent variables. Why this value is different from the Solow model? [8]
- d) Consider a closed economy and the government permanently raises government spending while financing it through lump-sum taxation. Explain how the phase diagram for capital and consumption changes. analyze whether a temporary increase in government purchases can have different welfare implications compared to a permanent increase. [6]

Tutorial exam 02

502
2

Time: 50 minutes; Marks 20

Consider a closed economy that follows the Solow neoclassical growth model with a Cobb–Douglas production function

$$Y(t) = K(t)^\alpha [A(t)L(t)]^{1-\alpha}, 0 < \alpha < 1$$

where labor $L(t)$ grows at rate n , technology $A(t)$ grows at rate g , and capital depreciates at rate δ . A constant fraction s of output is saved.

- a) Why is the Solow model referred to as a neoclassical growth model? Derive the dynamic equation of capital per effective worker and explain the conditions under which the economy converges to a steady state.
- b) Examine the transition process of capital per effective worker after a permanent decrease in the depreciation rate δ . How would a decline in population growth rate due to demographic transition affect Bangladesh's steady-state capital per worker?
- c) Use phase-diagram intuition to explain the adjustment path of capital per effective worker, output per effective worker, and consumption per effective worker.
- d) Suppose the economy initially operates at the Golden Rule level of capital. Examine how a permanent increase in the saving rate affects welfare over time. Is it possible for such a policy to improve long-run welfare? Justify your answer analytically.
- e) Critically evaluate the Solow model's prediction of conditional convergence. Discuss at least two reasons why empirical growth experiences of developing countries may deviate from this prediction.

[5+3+4+4+4]

Second Tutorial
Development economics (course No 503).
Department of economics, Jahangirnagar University
(Answer any ONE); Time 55 minutes. Marks: 20.

1. How do you explain Bangladesh's status as a 'positive outlier' in cross country comparisons of social progress?
2.
 - i. What is Classical Patriarchy?
 - ii. Please mention some of the different forms of women's agency.
 - iii. Given the context of Bangladesh, how do you explain the evolution of women's agency in its development trajectory?
3.
 - A. Explain the concepts below (according to the UN definition)
 - i. Persons in Employment
 - ii. Labor Underutilization
 - B. Are the Persons in own-use production work considered as 'employed'? Please provide an argument in favor of your answer.
 - C. What is the proportion of unemployed youth aged 15--29 in total unemployment in Bangladesh? How is the situation in terms of NEET with Youth?
4. From Venezuela's experience as an 'Oil Rich Country', how do you explain their investment and governance structure to take benefit from their own resource?

Department of Economics
Jahangirnagar University
Tutorial Examination
Labor Economics (Econ 510)
Marks: 20 Time 50 minutes

510
1

Answer the following questions:

1. a) Explain the forces that are responsible for labor market outcomes. 10
- b) Develop a two-way labor leisure model and determine the optimal amount of labor and leisure hour. 10

Quiz # 02

Course No: 510 (Labor Economics)
Full Marks: 20 Time: One Hour

Answer any one of the following questions:

1. a) Explain the assumptions of the Marginal Productivity Theory of Labor Demand. Are these assumptions valid in real world? How did the economists try to defend this theory? Briefly explain. 15
- b) What are the other factors than wages that influence the firm's level of employment, explain. 5
2. a) Explain the theory of labor demand in the long run. 5
- b) Explain the laws of derived demand. 10
- c) With the help of appropriate graph(s) explain the change in the labor demand in the short-run and in the long-run due to a change in the wage rate. 5

Exam 2, Econ 512
Total marks 15, Date: February 3, 2026

512
2

1. Why do you think the demand for meat is rising in the developing countries? (4) b. *Excessive buffalo production has negatively affected the agricultural land and food production in India.* Explain. (6) c. Bangladesh has seen a rise in Nepia grass production in the last one decade. How do you assess such rise in Grass production? (5)

OR

2. The WTO Agreement on Agriculture (AoA) provides a set of rules for international trade policies in agriculture with the aim of ensuring free and fair competition. However, the promise of free trade in agriculture has turned out to be very different from the reality on the ground. The negative impact of the agreement has raised huge controversy within the developing countries. It is claimed that this agreement eventually protects the financial interest of western industrial countries while jeopardizing the interests of the developing countries. Explain. (15)

512
1

Course- 512, Tutorial 1, October 28, 2025. Total marks: 15

Answer 1 of the 2 following questions:

1. a. Discuss your opinion on at least 2 major critiques of the export oriented development model. (4) b. Discuss a specific export oriented industry in Bangladesh which has largely altered the land, labor, and environmental pattern of a particular region of the country. (4) c. Can you draw a parallel incident from the colonial history of Africa, which showed very similar pattern? What does it tell about the continuation of international economic exploitation in the developing part of the world? (7)

১) ক. রপ্তানিমুখী উন্নয়ন মডেলের দুটি সমালোচনা নিয়ে তোমার মতামত ব্যাখ্যা করো। (৪) খ. বাংলাদেশের একটি রপ্তানিমুখী শিল্প নিয়ে আলোচনা করো যা ইতিমধ্যেই দেশের একটি অঞ্চলের জমি, পরিবেশ এবং শ্রমিক বিন্যাসে ব্যাপক পরিবর্তন এনেছে। (৪) গ. আফ্রিকা মহাদেশের উপনিবেশিক সময়ের প্রেক্ষাপটে একই ধরনের একটি অর্থনৈতিক ঘটনার তুলনামূলক আলোচনা করো। এই আলোচনা থেকে আন্তর্জাতিক অর্থনৈতিক শোষণ ব্যবস্থার পুনরাবৃত্তি উপর আলোকপাত করো। (৭)

OR

২. UNCTAD (The United Nations Conference on Trade and Development) report (2013) has suggested the developing countries to focus on domestic demand instead of 'western' demand.

a. What is the UNCTAD report's assessment on various risks associated with the external demand driven industrialization model in the developing countries? (4) b. Why did UNCTAD emphasize on the importance of stimulating domestic forces of demand in developing countries? (4) c. According to the UNCTAD report, both the state and the domestic private investors could play a crucial role in expanding domestic demand in developing countries. Explain How? (7)

২. জাতিসংঘের বাণিজ্য ও উন্নয়ন বিষয়ক সংস্থা (UNCTAD) তাদের ২০১৩ সালের প্রতিবেদনে উন্নয়নশীল দেশগুলোকে পশ্চিমা চাহিদার পরিবর্তে স্থানীয় চাহিদায় মনোযোগ দেয়ার পরামর্শ দিয়েছে। ক. উন্নয়নশীল দেশগুলোতে পশ্চিমা চাহিদা ভিত্তিক শিল্পায়নের ঝুঁকি বিষয়ে এই সংস্থার মূল্যায়ন কি? (৪) খ. উন্নয়নশীল দেশগুলোতে স্থানীয় চাহিদাকে উৎসাহিত করার পেছনে এই সংস্থার যুক্তিগুলো কি? (৪) গ. স্থানীয় চাহিদা বৃদ্ধিতে রাষ্ট্র ও স্থানীয় বিনিয়োগকারীদের ভূমিকা বিষয়ে তাদের পরামর্শ কি? (৭)

Jahangirnagar University
Department of Economics
1st Tutorial Exam
Course: Econ-513 (Contemporary Economic Thoughts)
Time: 1.00 Hour Marks: 20

1. Critically examine the contributions of Jean-Baptiste Say and Nassau William Senior to classical political economy. In your answer, discuss Say's Law of Markets and theory of entrepreneurship alongside Senior's abstinence theory of profit and views on capital, and evaluate how their ideas shaped the classical understanding of production, distribution, and the role of the capitalist.

OR

2. Critically examine the development of marginal utility theory from Jeremy Bentham's utilitarian philosophy to the formal marginal analysis of Jevons, Menger, and Walras. In your answer, compare their approaches to value, utility, and price determination, and evaluate how their contributions collectively marked the transition from classical political economy to the Marginal Revolution.

Jahangirnagar University
Department of Economics
2nd Tutorial Exam
Course: Econ-513 (Contemporary Economic Thoughts)
Time: 1.00 Hour Marks: 20

Part A: Alfred Marshall [Answer any One]

- 1) Explain Marshall's synthesis of classical political economy and marginal utility theory. How did this synthesis reshape modern microeconomic analysis?
- 2) Critically evaluate Marshall's theory of price determination. How does the interaction of demand, supply, and time-period analysis form the foundation of his economic system?

Part B: Karl Marx [Answer any One]

- 1) Explain Marx's theory of surplus value. How does it form the basis of his critique of capitalist exploitation?
- 2) Discuss Marx's theory of capital accumulation and crisis. Why did he consider capitalism inherently unstable?

TUTORIAL 3 (06052026)

Department of Economics, MSc (Econ 503). Time 50 minutes

(answer any ONE)

1. If you are in a position to present a brief overview of Bangladesh's macro-economic scenario, which indicators will be picked up and why? How do you present a brief macro-economic overview of Bangladesh? (05+15)
2. Please briefly unveil key challenges of the external trade in Bangladesh amidst current dynamic landscape.
3. Provide an overview how IRAN strategized its ODA dependency and use of its Energy resources over decades amidst global challenges? How did these crises demonstrate pressure on economic and social outcome? (10+10)

Overseas Development Assistance

Department of Economics
Jahangirnagar University

Quiz # 03

Course No: 510 (Labor Economics)
Full Marks: 20 Time: One Hour

Answer any one of the following questions:

1. a) What is Law of One Wage? Explain the mechanism of determining the equilibrium wage rate and number of labor in competitive markets. 6
b) In real world labor markets why does the "one wage" not exist? 6
c) Construct a realistic model for determining wages. 3
d) Explain the dual labor market theory. 5
2. a) What is human capital theory? In what forms can an individual invest in him/herself? 3
b) A high school graduate has two options entering in the labor market or pursue a four-year college degree. Both options have their benefits and cost. On what basis will the high school graduate make this decision? Explain briefly. 10
c) What are the implications of Human Capital Theory. 7

Exam 3, Econ 512

Date: May 9, 2026

Total marks 15

1. What led to the development of Modernization theory back in the 1940s and 50s (3)
Examine 4 critiques of Modernization theory. (12)

Consider a simplified version of the models of R&D and growth model developed by P. Romer (1990), Gross and Helpman (1991) and Aghion and Howitt (1992) where θ reflects the effect of the existing stock of knowledge on the success of R&D and B is a shift parameter. Both goods producing sector and R&D producing sector use the full stock of knowledge A .

- a) Critically explain the role of a_L (the labor force used in the R&D sector) for the dynamics of knowledge accumulation without considering capital into the model. Is there any long-run growth effect for change in a_L ? [6]
- b) Assume that $\theta + \beta < 1$ and $n > 0$, and that the economy is on its balanced growth path. Describe how each of the following changes affects the $\dot{g}_A = 0$ and $\dot{g}_K = 0$ lines and the position of the economy in (g_A, g_K) space at the moment of the change:

- (i) An increase in n .
- (ii) A decrease in a_K .

[6]

- c) There are two cases $\beta + \theta < 1$ or $\beta + \theta = 1$ and $n = 0$; What is the difference on the dynamics of the growth rates of capital and knowledge between these two cases? [4]

- d) Briefly explain the nature of Knowledge and the determinants of the allocation of resources to R & D. [4]

Answer any ONE

Time: 45 minutes

1. a) Define general equilibrium. In what respect it differs from the partial equilibrium.

Examine with examples.

8

- b) Graphically explain the problems of existence, uniqueness and stability in connection to general equilibrium.

12

2. a) Explain with assumptions of a $2 \times 2 \times 2$ model, the production equilibrium in a simple general equilibrium model through Edgeworth box diagram.

10

- b) Graphically explain how an external shock in the product market affects the factor market and the mechanism through which equilibrium is restored in both markets.

10

3. a) Define social welfare. Critically examine three key issues that Pigou dealt with in analyzing social welfare. (5)

10

- b) Derive the grand utility possibility frontier and determine the 'point of bliss'. Does such a point ensure maximum social welfare-justify?

10

4. a) 'Pareto optimality is a necessary but not sufficient condition for social welfare maximization' – justify the statement considering a $2 \times 2 \times 2$ model. — (2)

10

- b) Explain the underlying assumption about the marginal utility of money for being the Hicks –Kaldor compensation be correct? — (3)

10